

A gliding affine-slab argument for Banach-space well-separating common complements

Autonomous mathematical discovery run `fa_banach_001`
`agent_lane_00` — model GPT5.6

13 August 2026

Abstract

Noethen asked whether every sequence of closed subspaces of the same finite codimension in a real Banach space has a well-separating common complement, and whether the tuples spanning such complements are prevalent. We prove both conjectures. The missing codimension-one assertion follows from a translated finite-dimensional slab estimate and a summable gliding construction. Quantitatively, for every sequence (φ_j) of norm-one functionals on an arbitrary Banach space there is a unit vector x such that $|\varphi_j(x)| \geq \exp[-O((\log(j+1))^2)]$ for every j .

Status. This is a candidate new full proof of Conjectures 3.4 and 3.15 in [1]. A bounded search found no later resolution; no priority claim is made.

1 The source questions

Let X be a real Banach space. For closed subspaces $V_j \subset X$ of common finite codimension k , a k -dimensional common complement C is *well-separating* when there are $\delta_j > 0$ such that

$$\inf_{x \in C, \|x\|=1} \text{dist}(x, V_j) \geq \delta_j, \quad \lim_{j \rightarrow \infty} \frac{1}{j} \log \delta_j = 0.$$

Conjecture 3.4 of [1] asserts existence for every Banach space; Conjecture 3.15 asserts prevalence of the tuples spanning such complements. The source explicitly reduces existence to the Banach-space version of its Lemma 3.6, concerning a sequence of norm-one functionals. We prove a quantitative strengthening of that lemma.

2 A translated finite-batch estimate

Write B_2^n for the Euclidean unit ball and ω_n for its volume.

Lemma 1 (affine-slab avoidance). *Let X be a real Banach space, let $\varphi_1, \dots, \varphi_N \in X^*$ have norm one, and let $z \in X$ and $r > 0$. There is $h \in X$ such that*

$$\|h\| < r, \quad |\varphi_j(z+h)| \geq \frac{r}{8N^2} \quad (1 \leq j \leq N).$$

Proof. Put $K = \bigcap_{j=1}^N \ker \varphi_j$ and let $q : X \rightarrow Y := X/K$ be the quotient map. Then $n := \dim Y \leq N$, and each φ_j induces a functional $\psi_j \in Y^*$ of norm one. Indeed, this is the standard isometric identification of $K^\perp$ with $(X/K)^*$.

By John's ellipsoid theorem, choose a linear isomorphism $S : \mathbb{R}^n \rightarrow Y$ for which

$$S(B_2^n) \subset B_Y \subset \sqrt{n} S(B_2^n).$$

Let $\ell_j = \psi_j \circ S$ and $a_j = \|\ell_j\|_2$. Since $\|\psi_j\| = 1$ and $B_Y \subset \sqrt{n} S(B_2^n)$,

$$a_j \geq n^{-1/2}. \quad (1)$$

Set $R = r/2$ and choose U uniformly from RB_2^n . For any real b , an orthogonal change of coordinates and slicing perpendicular to ℓ_j give

$$\mathbb{P}(|\ell_j(U) + b| < \eta) \leq \frac{2\eta}{a_j R} \frac{\omega_{n-1}}{\omega_n}. \quad (2)$$

The estimate is uniform in the translate b : the relevant interval has length at most $2\eta/a_j$, and every ball section has volume at most $\omega_{n-1}R^{n-1}$. Also $\omega_{n-1}/\omega_n \leq \sqrt{n}$. For $n \geq 2$, this follows by slicing the unit ball and integrating over $|t| \leq n^{-1/2}$, since $2(1 - 1/n)^{(n-1)/2} \geq 1$; the case $n = 1$ is immediate. Together with (1), (2) becomes

$$\mathbb{P}(|\ell_j(U) + b| < \eta) \leq \frac{4\eta n}{r}.$$

Take $b = \varphi_j(z)$ and $\eta = r/(8N^2)$. A union bound shows that the probability that at least one of the N inequalities fails is at most

$$N \frac{4\eta n}{r} = \frac{n}{2N} \leq \frac{1}{2}.$$

Hence some $u \in RB_2^n$ satisfies $|\psi_j(qz + Su)| \geq \eta$ for all j . The quotient vector $y = Su$ has $\|y\|_Y \leq r/2$. By the definition of the quotient norm it has a lift $h \in X$ with $q(h) = y$ and $\|h\| < \|y\|_Y + r/2 \leq r$. This lift has all the required properties. $\square$

3 One vector for the whole sequence

Theorem 2 (Banach version of Noethen's Lemma 3.6). *Let X be an arbitrary real Banach space and let $(\varphi_j)_{j \geq 1} \subset X^*$ be norm-one functionals. There are a unit vector $u \in X$ and positive numbers δ_j such that*

$$|\varphi_j(u)| \geq \delta_j \quad (j \geq 1), \quad \lim_{j \rightarrow \infty} \frac{1}{j} \log \delta_j = 0.$$

More precisely, one may take

$$\delta_j = \frac{3}{4}, 2^{-(m(j)^2 + 7m(j) - 1)}, \quad m(j) = \max\{1, \lceil \log_2 j \rceil\}.$$

Proof. Let $M_m = 2^m$ and $\rho_1 = 1/4$. Recursively set

$$\alpha_m = \frac{\rho_m}{8M_m^2}, \quad \rho_{m+1} = \frac{\alpha_m}{8}.$$

Starting with $x_0 = 0$, apply Lemma 1 at stage m to the base point x_{m-1} , the first M_m functionals, and radius ρ_m . It supplies h_m with $\|h_m\| < \rho_m$ such that, for $x_m = x_{m-1} + h_m$,

$$|\varphi_j(x_m)| \geq \alpha_m \quad (1 \leq j \leq M_m). \quad (3)$$

Since

$$\frac{\rho_{m+1}}{\rho_m} = \frac{1}{64M_m^2} \leq \frac{1}{256},$$

the series $x = \sum_{m \geq 1} h_m$ converges and

$$\|x\| < \sum_{m \geq 1} \rho_m \leq \frac{1/4}{1 - 1/256} = \frac{64}{255} < 1.$$

Moreover, the tail after stage m satisfies

$$\sum_{k > m} \rho_k \leq \frac{\rho_{m+1}}{1 - 1/256} = \frac{32}{255} \alpha_m < \frac{1}{4} \alpha_m. \quad (4)$$

Thus (3) and (4) imply

$$|\varphi_j(x)| > \frac{3}{4} \alpha_m \quad (1 \leq j \leq M_m). \quad (5)$$

The recurrence is explicit:

$$\alpha_1 = 2^{-7}, \quad \alpha_{m+1} = \alpha_m 2^{-(2m+8)}, \quad \alpha_m = 2^{-(m^2+7m-1)}.$$

For $m = m(j)$, inequality (5) gives the stated δ_j . If $2^{m-1} < j \leq 2^m$, then $-\log \delta_j = O(m^2) = O((\log j)^2)$, so $j^{-1} \log \delta_j \rightarrow 0$.

Finally, (5) for $j = 1$ shows that $x \neq 0$. Put $u = x/\|x\|$. Since $\|x\| < 1$, normalization only increases all the absolute functional values. $\square$

Remark 3. *No separability of X is used: each correction lives over a finite-dimensional quotient, and the corrections converge in X itself. For a complex Banach space, applying the theorem to the real norm-one functionals $\operatorname{Re} \varphi_j$ also gives $|\varphi_j(u)| \geq |\operatorname{Re} \varphi_j(u)|$, so the same conclusion holds.*

4 Resolution of both conjectures

Corollary 4. *Conjectures 3.4 and 3.15 of [1] are true. Namely, every sequence of closed subspaces of a real Banach space having the same finite codimension has a well-separating common complement, and the tuples spanning such complements form a prevalent subset of the corresponding finite Cartesian power of the space.*

Proof. First suppose the subspaces V_j are hyperplanes. Choose $\varphi_j \in X^*$ with $\|\varphi_j\| = 1$ and $V_j = \ker \varphi_j$. For the unit vector u from Theorem 2,

$$\operatorname{dist}(u, V_j) = |\varphi_j(u)| \geq \delta_j.$$

Consequently $\operatorname{span}(u)$ is a well-separating common complement.

For completeness, the arbitrary-codimension propagation in [1] is purely Banach-space theoretic. Inductively, for subspaces of codimension $k+1$, enlarge each V_j to a codimension- k subspace and obtain a well-separating k -dimensional complement C_1 . Then $V_j \oplus C_1$ is a hyperplane, so the codimension-one result supplies a common complementary line C_2 . Lemma 3.8 of the source, applied in X/V_j , shows that $C_1 \oplus C_2$ has a transversality lower bound

$$\frac{1}{2\sqrt{5}} \delta_j^{(1)} \delta_j^{(2)}.$$

The product of two subexponentially decaying positive sequences is again subexponential. This proves Conjecture 3.4 for every finite codimension.

Finally, Proposition 3.20 of [1] proves in an arbitrary Banach space that the existence of one well-separating common complement makes the set of all spanning tuples prevalent. Applying it to the complement just constructed proves Conjecture 3.15. $\square$

Verification notes

The proof uses only John’s ellipsoid theorem (already recorded as Theorem 2.5 in the source), elementary Euclidean slicing, quotient norms, and the two Banach-space propagation results proved in the source. The decisive feature of Lemma 1 is uniformity in the affine translate z ; this lets successive finite-batch solutions be added without losing earlier margins. The source conjecture pages and a separate constant-and-tail audit accompany this packet.

References

- [1] F. Noethen, *Well-separating common complements of a sequence of subspaces of the same codimension in a Hilbert space are generic*, arXiv:1906.08514 (2019; PDF revision dated 9 March 2022).